

Assignment 04
Third Year BS (Honors) 2024-2025
Course Title: Math Lab III, Course Code: AMTH 350
Department of Applied Mathematics, University of Dhaka

Name:

Roll No.:

Group:

Day 1

1. Estimated values of thermal conductivity of silicon at different temperatures are given in the following table.

$T(K)$	2	4	6	8	10	20	40	60
$k(W/m-K)$	46	300	820	1560	2300	5000	3500	2100
$T(K)$	80	100	150	250	350	500	1000	1400
$k(W/m-K)$	1350	900	400	190	120	75	30	20

- (a) Make a plot of k versus T using log scale on both axes.
 (b) Curve-fit the data with a second-order polynomial $y = ax^2 + bx + c$ in which $x = \log(T)$ and $y = \log(k)$. Once the coefficients a , b , and c are determined, write an equation for k as a function of $\log(T)$. Use this equation for curve-fitting the data. Make a second plot that shows the data points with markers and the curve-fitted equation with a solid line.
 (c) Repeat part (b) using a third-order polynomial.
2. The standard air density, D (average of measurements made), at different heights, h , from sea level up to a height of 33 km is given below.

$h (km)$	0	3	6	9	12	15
$D (kg/m^3)$	1.2	0.91	0.66	0.47	0.31	0.19
$h (km)$	18	21	24	27	30	33
$D (kg/m^3)$	0.12	0.075	0.046	0.029	0.018	0.011

- (a) Make the following four plots of the data points (density as a function of height): (i) both axes with linear scale; (ii) h with log axis, D with linear axis; (iii) h with linear axis, D with log axis; (iv) both log axes. According to the plots, choose a function (linear, power, exponential, or logarithmic) that best fits the data points and determine the coefficients of the function.
 (b) Plot the function and the points using linear axes.
3. Molecules of a gas in a container are moving around at different speeds. Maxwell's speed distribution law gives the probability distribution $P(v)$ as a function of temperature and speed:

$$P(v) = 4\pi \left(\frac{M}{2\pi RT} \right)^{3/2} v^2 e^{-Mv^2/(2RT)}$$

where M is the molar mass of the gas in kg/mol, $R = 8.31J/(mol K)$, is the gas constant, T is the temperature in kelvins, and v is the molecule's speed in m/s.

Write a MATLAB code that draws the distribution function $P(v)$ versus v for $0 < v \text{ m/s} < 3000$. Make several graphs in the same plot for different values of temperature (at least four). Label the axes and display a legend. The most probable speed, v_p , is the speed most likely to be possessed by any molecule in the system and corresponds to the maximum value or the mode

of $f(v)$. Find the most probable speed for each particular temperature and draw a curve or trend line by taking these points.

Day 2

4. The relationship between the population of lions (predators), N_L , and the population of gazelles (prey), N_G that reside in the same area can be modelled by a system of two ODEs. Suppose a community consists of N_L lions (predators) and N_G gazelles (prey), with b and d representing the birth and death rates of the respective species. The rate of change (growth or decay) of the lion (L) and gazelle populations can be modelled by the equations:

$$\frac{dN_L}{dt} = b_L N_L N_G - d_L N_L$$

$$\frac{dN_G}{dt} = b_G N_G - d_G N_L N_G$$

at $t = 0$, $N_G = 3000$, and $N_L = 500$. The coefficients in the model are: $b_G = 1.1 \text{ yr}^{-1}$, $b_L = 0.00025 \text{ yr}^{-1}$, $d_G = 0.0005 \text{ yr}^{-1}$, and $d_L = 0.7 \text{ yr}^{-1}$.

Determine the population of the lions and gazelles as a function of time from $t = 0$ to $t = 25$ years by using (a) modified Euler's method, (b) Taylors method of order 4, (c) Runge-Kutta method of order 4, (d) multistep Predictor- Corrector method and (e) MATLAB default solver ode45. Make a table whose 1st column will be values of t , and other columns will be results obtained by different numerical methods. Also make a plot of the obtained data in the same window with different line style and colour directives.

5. The Foucault pendulum, which is free to swing in both the x and y directions, is often used to demonstrate the earth's rotation. Its equations of motion (in the absence of damping) are described by:

$$\frac{d^2 x}{dt^2} - (2\omega \sin(l_a)) \frac{dy}{dt} + \frac{g}{L} x = 0$$

$$\frac{d^2 y}{dt^2} - (2\omega \sin(l_a)) \frac{dx}{dt} + \frac{g}{L} y = 0$$

where $\omega = 7.29 \times 10^{-5} \text{ rad/s}$ is the angular velocity of the earth's rotation, l_a is the latitude of the pendulum location in degrees, $g = 9.81 \text{ m/s}^2$ is the acceleration due to gravity, and $L = 10 \text{ m}$ is the length of the pendulum. Solve the system of equations and determine how long it would take for the plane of swing of the pendulum to rotate by 45° . Use $l_a = 39.9611^\circ$ (for Columbus, Ohio), and the following initial conditions:

$$x(0) = 1, \quad \left. \frac{dx}{dt} \right|_{t=0} = 0, \quad y(0) = 0 \quad \text{and} \quad \left. \frac{dy}{dt} \right|_{t=0} = 0$$

Reduce the system of two second-order ODEs to a system of four first-order ODEs and solve using methods mentioned in the previous problem as well as MATLAB's built-in functions.